Bedform Finder Software

User Manual

Data Importing, Multibeam Bathymetric Processing, and Data Visualization

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INTRODUCTION

Bedform Finder is a software app developed for importing, processing, analyzing, and visualizing multibeam bathymetric survey data for automated bedform analysis. The software integrates data acquisition, bed elevation profile (BEP) generation, peak and trough detection, visualization, grouping, and manual quality control into a unified workflow.

The software was designed to improve the efficiency and reproducibility of bathymetric bedform analysis. Bedform Finder supports both individual survey analysis and large-scale batch processing workflows and provides tools for interactive visualization and quality assurance.

This manual describes the workflow, inputs and outputs, processing methods, visualization tools, grouping functionality, quality control procedures, line creation procedures, performance considerations, and software maintenance practices associated with Bedform Finder.

1) WORKFLOW

1.1) ARCHITECTURE

The software separates data import, processing, visualization, grouping, and quality control into interconnected components which may operate independently or as part of a complete end-to-end workflow.

1.1.1) Recommended Workflow

- Begin processing with raw multibeam bathymetric data. This either can be obtained via the built in eHydro tool or from another source.
- Additionally, a line file is required. This can be obtained through the built-in line creation tool, use a preloaded line from the application dependencies, or from another source
- Once the data and the line are obtained, process the dataset in the single processor module or multiple datasets in the batch processor module.
- If the data results have quality issues, use the built-in masker tool to mask away areas such as sandbars, scour holes, or other erroneous data.
- Use the single plot tools to visualize the output results for the dataset.
- If multiple datasets were processed and would like to be compared, use the built-in grouping tool to group together individual datasets.
- Once grouped, use the group plot tools to visualize the results of groups.

1.2) PARAMETERS

Bedform Finder requires a relatively small set of user-defined parameters for bathymetric processing. These parameters control the eHydro tool, bed elevation profile generation, filtering behavior, and peak-trough detection sensitivity.

1.2.1) eHydro Tool

The eHydro tool has many inputs including start and end date, river, quantity, xmin, ymin, xmax, and ymax.

- Start date defines the beginning date that surveys will be searched for in the eHydro database.
- End date defines the ending date that surveys will be searched for in the eHydro database
- River or River code defines which river the user would like to search for. The software receives all results from the eHydro API and filters the desired river by the beginning letters of the filename. The river dropdown provides some common river codes.
- Quantity dictates the number of surveys that will be downloaded. This number is only considered after the fetch when a download has been initiated. The all button will download all the surveys that were found in a fetch.
- Xmin, ymin, xmax, and ymax define a bounding box on the eHydro map. This bounding box is drawn over the continental US in red. By changing the values for the bounding box, the map is updated accordingly. The values for the bounding box are to be given in geospatial coordinates.

1.2.2) Single Processor Tool

The single processor includes inputs for the line file, data file, input CRS or EPSG code, vertical units, vertical type, offset distance, sample spacing, prominence, high pass filter threshold, and file names for data and figures.

- Line file is simply a file containing eastings and northings that give the predominant flow direction along a river. It can be an entire river or just a small reach of the river. It can either be created using the built-in line creation tool, obtained via preloaded files in the dropdown, or provided by the user. More information of the file requirements are given below in 2.1.2.
- Data file is simply a file containing the eastings, northings, and vertical info for a dataset of multibeam bathymetric data. More information of the file requirements are given below in 2.1.1.

- Input CRS or EPSG code defines the coordinate reference system for a dataset and a line. If the user creates or provides a river line file, it is naively assumed that it follows the same coordinate system as the data. When using a line obtained via the dropdown menu, this reference system only needs to match the data. A variety of common coordinate reference systems on the eHydro website are available in the dropdown menu. If the eHydro tool was used, the eHydro option should be selected as all data was projected to EPSG 5070, or the Contiguous Albers Equal Area conic projection. For other data, any worldwide projection system can be entered by giving the valid corresponding EPSG code.
- Vertical Units define the units of the vertical data. This is simply either in meters or US survey feet. Any other units must be converted using another program before processing in the Bedform Finder software.
- Vertical Type defines whether the data is given in depth or meters. Most multibeam bathymetric sets are given in depth, however the option remains available for elevation given any edge cases. This simply inverts the data by multiplying by minus one for depth.
- Offset Distance gives the transverse distance between parallel Bed Elevation Profiles (BEPs). Auto will by default use a value of 1 meter.
- Sample Spacing gives the distance between points that lie on each BEP. Auto by default uses a resolution of 1 meter due to the low resolution of multibeam bathymetric data.
- Prominence is a parameter that directly affects the peak and trough detection. It defines the minimum height from trough to peak that must be achieved for the algorithm to consider a peak. Auto will calculate a prominence value. Use with caution as this has not been verified.
- High Pass Filter Threshold is a parameter that directly affects the peak and trough detection. It defines the maximum wavelength in each BEP. Larger wavelengths are removed using the continuous wavelet transform. Auto will set by default a value of 100 meters.
- Filenames for both data and figures are not necessary for saving. If the field is left empty, the filename of the input data will be used to build a unique filename for the output data

1.2.3) Batch Processor Tool

The batch processor includes the exact same set of parameters as the single processor, however, it differs slightly. The user must select between running an

entire folder or selecting only certain files from the folder. Additionally, the original filename of the input data is used for the output data.

- File Selection is done by choosing “Files” and then using the control or shift key to highlight multiple files. Once finished, select open. Reopening the file selection window clears all previous selections.
- Folder Selection is done by choosing the desired folder. All valid file types within the folder will be added to the queue.

2) IO

2.1) INPUTS

2.1.1) Data

Data can be obtained through two different methods. First is by download using the eHydro tool. Second is through providing a custom dataset.

When providing a dataset, the input data must consist of three columns: X, Y, and Z. Order matters. X and Y are given as eastings and northings while Z is an arbitrary vertical reference. Several file formats are acceptable including .XYZ, .CSV, .XLS, .XLSX, .TXT, or .MAT.

2.1.2) River Line

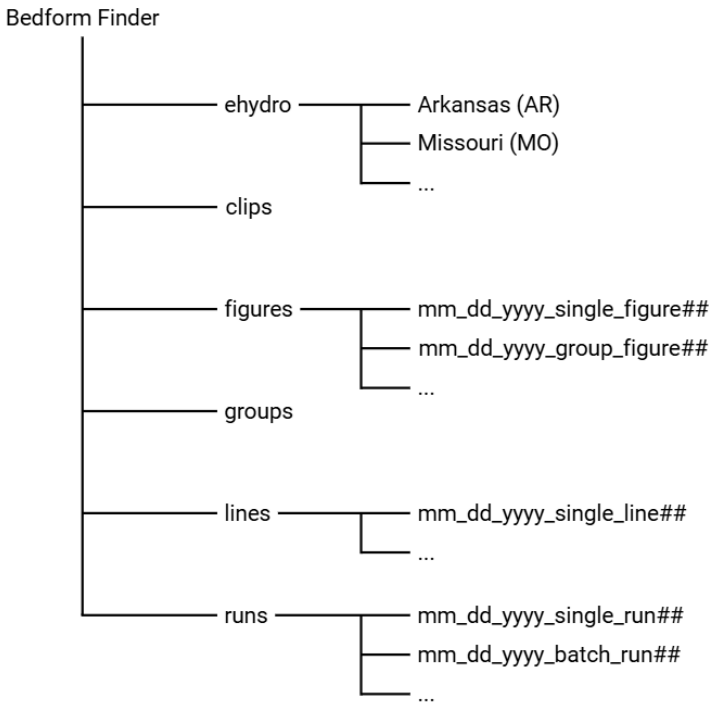
The River Line can be obtained through three different methods. The first is by using the Missouri River sailing line provided by the United States Army Corps of Engineers. Second is by creating a custom line using the built-in line maker tool. Third is by providing a custom dataset of points that make up a river line.

When providing a custom line, the input river line must consist of at least two columns: X and Y. Optionally, a third column RM can be included. Order matters. X and Y again are given as eastings and northings while RM is the river mile at each point. Several file formats are acceptable including .XYZ, .CSV, .XLS, .XLSX, .TXT, or .MAT. It is recommended that the resolution of any line used meet or exceed a sample spacing of 0.1 miles of roughly 200 meters.

2.2) OUTPUTS

2.2.1) File Structure

A file structure is automatically generated and adheres to a set of rules. Upon launch, Bedform Finder preferentially chooses the documents folder; it tries the personal one drive, commercial one drive, and will settle on the user. A new folder “Bedform Finder” is created as the root. This root can be changed in the settings.



As shown in the image above, subfolders will be created in the root. The eHydro tool creates a subfolder in the root called “ehydro” where folders are further subdivided based upon river code. The processing tools both single and batch save into a subfolder called “runs” where folders are further subdivided for each single or batch process. The grouper tool creates a subfolder in the root called “groups” where each saved group is added. The masker tool saves to a subfolder in the root called “clips” where each data file is added. All visualizations save to a subfolder called “figures” where folders are subdivided. Finally, all lines created in the line maker tool get saved to a subfolder called “lines” where another folder is created for each line.

2.2.2) Data

The output data structure as shown below contains information on the raw bathymetry data, processed bedform dimensions, and other metadata. There are three supported file formats for the output data. First is .MAT, which prioritizes speed and size. However, this format is not open source and not readable without a MATLAB license. Second is .JSON which prioritizes readability. Using a text or code editor you may even manually change values. This format is very large due to storing plain text. Finally is .HDF5, which is an open-source file format. This format is not readable however it is a good compromise between .JSON and .MAT. It is recommended that as long as data does not need to be extracted, use the .MAT file format. Below is an image of the fields contained within the output data structure.



2.2.3) Figures

Figures support a variety of output formats, including some compressed image types like .PNG and .JPG, vector graphics like .PDF and .EPS, raw pixel data from .TIFF and .BMP. Also available for saving is the .FIG format which saves the MATLAB figure handle with all contained data. The one downside to this format is it does require a MATLAB license to view. Use visualization tools built into Bedform Finder to replot data for the interactive viewing. Types of plots are discussed in 5.1 and 5.2.

2.2.4) eHydro

The built in eHydro downloading tool outputs .CSV files containing three delimited columns of data: X, Y, and Z. Additionally, eHydro keeps track of a file named .processed. This file keeps track of all surveys in the directory. Not only downloaded, but also files that were checked but did not meet the criteria. This way files that were not downloaded due to failing the size and density criteria are not reprocessed in the future again. The file size test simply requires the file to be larger than 1000kB. Density finds the total number of filled cells compared to cells that have greater than 100 points of data. The cell size has been set at 100 meters, meaning most surveys have greater than 8 blocks. This ratio must exceed 0.2. The .processed is plain text so the user can remove or add filenames with a text editor.

3) PROCESSING

3.1) EHYDRO

To start downloading with eHydro follow this set of steps:

3.1.1) Guide

- Enter in the parameters for start date, end date, river code, and the bounding box limits. Use the mouse to zoom and pan around the map of the United States.
- Select fetch to query the eHydro API and obtain the list of surveys to download.
- Once the fetch is complete, enter the number of surveys to download and the folder location to download to. The recommendation is to leave the folder location empty and set a default location in the settings menu.
- Click download to begin the download process
- If necessary, use the pause and stop buttons to halt the download process.

3.2) PROCESSING

Start processing either single or batch with this set of steps:

- Enter in the parameters for data, river line, EPSG code, vertical units and type, offset distance, sample spacing, prominence and high pass filter threshold.

3.2.1) Single

- Select compute to process the data set.
- Select a download folder location, create filenames, and choose an extension for the output data or figures. It is recommended to leave the folder location and filename empty. Set a default extension type and the default save directory in the settings.

- Select save data or save figures to save.

3.2.2) Batch

- Choose a download folder location and an extension type. It is recommended to leave the folder location empty. Set a default extension type and the default save directory in the settings.
- Select run to process all the data sets.
- To run batch with parallel processes, enable parpool in the settings menu. Set the number of workers. This is discussed further in the performance section.
- When running a batch process with parallel processing, stopping the operation is not possible. The stop button is only available for processes with no parpool.

3.3) GROUPING

Start grouping individual data with this set of steps:

3.3.1) Guide

- Begin grouping by adding files. Files will be added to the list box where they can be viewed.
- If files must be removed, select all files by holding the control key and selecting them. If the user would like to deselect, click the clear button and all selections will be removed. Once selected, use the remove selected button to remove those from the list.
- Once all files to be grouped have been added to the list, choose a filename, save directory and save extension. It is recommended to leave the folder location empty. Set a default extension type and the default save directory in the settings.
- Once ready click the group button to begin grouping. Files will be imported, appended to a group structure, and then exported to be used.

3.4) VISUALIZING

3.4.1) Single

- Select a processed individual or group set to view.
- Individual sets can be instantly plotted. Group sets can be plotted however use the group index to control which dataset within to plot.
- Single Plot types:
    - Length PDF:
        - Plots the probability density function histogram for dune length values. Plotted on top is the lognormal PDF of the data to compare.

- Height PDF:
  - Plots the probability density function histogram for dune height values. Plotted on top is the lognormal PDF of the data to compare.
- Steepness PDF:
  - Plots the probability density function histogram for dune steepness values. Plotted on top is the lognormal PDF of the data to compare.
- Length Heatmap:
  - Plots the length value located at each dune on a map to show spatially how the lengths are distributed. Also included is the boundary box of the raw data. This allows for comparison of the located bedforms and the raw data footprint.
- Height Heatmap:
  - Plots the height value located at each dune on a map to show spatially how the heights are distributed. Also included is the boundary box of the raw data. This allows for comparison of the located bedforms and the raw data footprint.
- Steepness Heatmap:
  - Plots the steepness value located at each dune on a map to show spatially how the steepness values are distributed. Also included is the boundary box of the raw data. This allows for comparison of the located bedforms and the raw data footprint.
- Peaks on Raster:
  - Plots a raster of the fluctuating bathymetry around the mean. low points are shown in blue and high points are shown in red. Plotted on top are the dune peaks in green. Generally, these should line up with the higher red areas, however, there are limitations to viewing the fluctuating bathymetry around the mean.
- Bed Elevation Profile:
  - Plots the individual profile pulled from a dataset. Located peaks and troughs are plotted, where peaks are red and troughs are blue.
  - Any data gaps will have the lagging trough removed; this is not an error and only a result of the indexing operation. Dune scales are not affected by this. It is purely visual.
  - To plot the Bed Elevation Profile (BEP), the profile index must be provided. The field to the right of the plotting button allows the user to alter this index.

- Raw Bathymetry:
  - Plots a raster of the raw bathymetry. The coordinates are given by whichever UTM coordinate reference system was assigned at processing. This can be found in the section on data 2.2.2.

3.4.2) Group

- Select a group set to view.
- Group Plot types:
  - Reach Location vs. Date:
    - Plots all the reach locations against the date to compare the spatial and temporal aspects of the data.
    - For this plot to work, there must be metadata available. See section 2.2.2 on how this metadata is added based on the filename of origin.
  - Length Scale Relation:
    - Plots the mean dune length of a reach vs the mean flow depth of that reach. Also included are the B&V 2017 scaling relationship and confidence intervals.
  - Height Scale Relation:
    - Plots the mean dune height of a reach vs the mean flow depth of that reach. Also included are the B&V 2017 scaling relationship and confidence intervals.
  - Steepness Scale Relation:
    - Plots the mean dune steepness of a reach vs the mean flow depth of that reach.
  - Height vs. Length:
    - Plots the mean dune height vs the mean dune length of each reach. Each point is color by the flow depth. Also included is the common height vs length scaling relationship.

3.4.3) Saving

Both Single and Group figures can be exported and saved. Set the save directory, filename, and file extension. Then select save and all open figures will be saved. It is recommended to leave the folder location empty. Set a default extension type and the default save directory in the settings.

3.5) QUALITY CONTROL

Start masking with this set of steps:

3.5.1) Guide

- Select a processed individual or group set to clip.
- Individual sets can be instantly clipped. Group sets can be clipped however use the group index to control which dataset within to reference. Once clipped and saved it will become an individual file.
- Once a dataset has been selected, generate the peaks on raster. Zoom and Pan around the map with the mouse to find areas to mask away.
- The clear plot button will reset the page entirely.
- To add masks, select the add mask button and begin drawing points. Encompass the area of peaks desired to remove. Either click on the beginning point or right click to complete the shape. Add more masks using the same process.
- To remove masks, right click on the mask and select delete.
- Once all desired masks are drawn, select clip. The clipped dataset will be regenerated on the plot.
- To save the clipped set, select save. It will be saved automatically in the default save directory specified in settings with the original filename and extension.

3.6) LINE CREATION

Start creating a custom line with this set of steps:

3.6.1) Guide

- Pick a set of raw data containing X, Y, and Z. More information on proper data can be found above in 2.1.1.
- Generate the raster. This may take a moment depending on the size of the dataset.
- One generated the software will allow the user to draw a line. Select draw line and begin placing points.
- It is recommended that greater than 6-8 points are used. Make sure that the line completely and fully pierces the dataset so that at least one point is outside of the data footprint.
- Once complete, double click to place the last point or press the right mouse button or the enter key to finish the line.
- To redraw the line, press the draw line button and redraw as specified above.
- Once complete, select save. The line will be saved with the same filename as the raw data in an excel file. It will be placed in the specified save directory. It is recommended to leave the folder location empty. Set a default save directory in the settings.

4) PERFORMANCE

Multibeam bathymetric datasets frequently contain millions of spatial points, making computational efficiency an important consideration within Bedform Finder. Several optimizations are incorporated to improve processing speed, reduce memory usage, and support large-scale batch analysis workflows.

4.1) PARAMETERS

4.1.1) eHydro

The greatest effect of the eHydro parameters is the bounding box and date range size. The algorithm performs a search recursively so larger bounding boxes and larger date ranges may take slightly longer to query. This is not significant however.

4.1.2) Processing

The greatest effect of the processing parameters is the original data file size and the BEP offset and sample spacing parameters. The longer a dataset is, the faster that the computation time will grow. Both interpolation and wavelet operations are nonlinear resulting in computation times that increase quickly. BEP offset and sample spacing both increase the number of operations to complete and the length of those operations respectively. Be mindful when setting resolution beyond what the data requires.

4.1.3) Visualizing

The greatest effect of the Visualizing parameters is the size of the raw data. Larger and denser multibeam datasets require more computations and therefore figures may take longer to generate.

4.2) BATCH PROCESSING

When batch processing, parallelization can provide a huge performance increase. To enable parallelization, make sure it is enabled in the settings. Then set the desired number of workers. This is maximized at 12 for all systems and a number is calculated based on the components of the system Bedform Finder is being run on. It is recommended that ¾ of the maximum is used as this is optimal. Parallelization can bring huge performance gains but setting the worker count too high can be slower due to initialization overhead and thread competition.

4.3) MISC

Some additional miscellaneous performance considerations must also be taken into account.

- When referring to the eHydro tool downloading, it is most greatly effected by download speeds and therefore faster internet will result in faster speeds. eHydro tool processing will not have a significant impact on speed.
- When exporting figure files setting the picture DPI too high will cause figure exporting to be slow. 300-600 is recommended. Further improvements on this speed can be made using a vector format such as .PDF
- When exporting data files, .JSON while human readable is not great for performance. If that data must be read, use .JSON. If not, it is recommended to use .MAT format as this is the fastest and smallest file format.

5) TROUBLESHOOTING

This section provides general troubleshooting guidance for common issues encountered while using Bedform Finder. Many problems arise from incorrect coordinate reference systems, improperly formatted input data, invalid river line geometry, or parameter selections incompatible with the spatial scale of the bathymetric dataset.

5.1) FIRST STEPS

Before reporting a bug or unexpected behavior, attempt the following steps:

- Restart the application:
  - Temporary issues related to memory usage, figure rendering, or file access may be resolved by restarting the software.
- Verify input data formatting:
  - Ensure bathymetric data contains valid numeric columns for eastings, northings, and depth/elevation values.
  - Verify river line files contain valid coordinate data.
  - Confirm no invalid text headers or unsupported formatting remain in the files.
- Confirm coordinate reference systems:
  - Verify that the correct EPSG code has been supplied.
  - Ensure bathymetric datasets and river lines use the same coordinate reference system.
- Try a different dataset
  - Running the workflow on a separate dataset can help determine whether the issue is data-specific or software-related.
- Reduce processing complexity
  - Try larger sample spacing values.
  - Reduce the number of files in batch processing.

    - Use smaller subsets of data to isolate problematic regions.
- Verify parameter selections
    - Excessively small sample spacing or offset distances may increase processing time and memory usage.
    - Improper prominence or filtering thresholds may produce poor peak-trough detection results.
- Refer to this manual
    - Many common issues are addressed within the workflow, IO, processing, visualization, and quality control sections of this manual.

5.2) REPORTING BUGS

If issues persist after completing the recommended troubleshooting steps, bugs or unexpected behavior may be reported through either the GitHub repository or direct email contact.

GitHub Repository:
https://github.com/MatthewRFree/BedformFinder

The GitHub repository may be used to:

- Report bugs
- Request features
- Track software updates
- Access source code and documentation

*NOTE: Code updates may be infrequent and code maintenance through github will be limited. For serious bugs please reach out through email

Direct Contact Information:

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- William Otero (United States Army Corps of Engineers)
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When possible, include example datasets or other information to assist with debugging. When reporting bugs, please provide:

- A description of the issue
- Steps required to reproduce the issue
- Input file formats used
- Processing parameters
- Error messages or screenshots